

Mechanical Property Characterisation of T300/914 Unidirectional Carbon Fibre Epoxy Matrix Composites

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Keywords
Carbon fibre
Polymer-matrix composites
(PMCs)
Material properties

Despite decades of development by Crag, BSI, ASTM, and ISO, differences in fixture design, specimen geometry, data-reduction methods, and anisotropic failure behaviour continue to influence reported properties. These effects are particularly pronounced in UD laminates, where high axial stiffness, weak transverse response, and direction-dependent failure mechanisms challenge the assumptions embedded within many standardised procedures. This study presents a comparative evaluation of tensile, compressive, shear, flexural, and fracture-toughness test methods applied to T300/914 carbon-epoxy laminates. The findings demonstrate that while many standards provide consistent results across material systems, there remains multiple factors that need to be considered prior to implementing and validating numerical models.

1.0 Introduction

Designing structures from polymer-matrix composites (PMCs) demands careful consideration at every stage, from the earliest analytical sizing calculations to the final qualification of full-scale components. The foundation of this process begins in the laboratory with coupon-level mechanical testing, where properties such as tensile, compressive, shear, flexural, and fracture toughness are established. During the 1980s and 1990s, significant efforts were devoted to standardising these measurements, with bodies like the Composites Research Advisory Group (CRAG) playing a pivotal role in consolidating and shaping recommendations that were proposed to the British Standards Institution (BSI). Through two decades of development, the insights gathered have supported the maturation of composite materials and have led to the development of international standards (1).

Although these standards have brought much-needed consistency, the practical balance between accuracy, repeatability, and ease of implementation has led to differing approaches across various testing approaches. Variations in data-reduction methods, fixture design, specimen geometry, and environmental or manufacturing conditions all contribute to discrepancies in reported properties (1). In addition, testing unidirectional composite materials presents a unique set of challenges that stem directly from their anisotropic nature. Unlike isotropic metals, where mechanical behaviour is uniform in all directions, unidirectional (UD) fibre composites are formed of lamina that exhibit extreme directionality in elastic modulus and strength. In turn, this also results in complex

failure mechanisms that should be considered when analysing experimental data.

These challenges become immediately apparent in tensile testing, where the high axial stiffness and weak transverse stiffness require careful experimental setup. As noted by Matthews in (1: p.48), since its introduction, the Crag RAE Method 300 standard remains one of the few standards dedicated to investigating the tensile behaviour of UD PMCs, with strict definitions on dimensions and load speed, unlike other standards like ASTM D3039M, which covers a wider scope for all composite materials that are symmetrical in the lamination direction. End crushing or “brooming” is a phenomenon that plagued early compression testing of UD composites due to pure end loading (ASTM D695). Over time, two other classes of compression standards have arisen in pure shear loading (Celanese fixture), and the combined loading through end and shear loading through the ICSTM methodology that was first introduced in 1993 by Haberle and Matthews (2). This fixture was designed to solve the problem of ensuring perfect alignment of the specimen in the Celanese jig employed in ASTM D3410M and Crag 400, while avoiding end crushing.

Shear characterisation of unidirectional composites can be done with multiple methods and standards established. Iosipescu (ASTM D5379M), $\pm 45^\circ$ tension test (ASTM D3518M), and short beam shear (ASTM D2344M) are of particular interest due to the ability to obtain both shear modulus and strength with relatively simple data reduction methods, making it a particularly attractive when specialist layup and machining expertise is available.

On the flip side, standards for investigating the Mode I fracture toughness (G_{IC}) and flexural properties of UD laminates have seen less debate. G_{IC} is often determined using the double

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cantilever beam (DCB) test (3), as outlined in the ASTM D5528, with correction factors to be calculated and counted for. Whereas ASTM D790 and more recently, ASTM D6272 have been well established for three-point and four-point flexural testing respectively. While it only provides a baseline and should only be used for quality checks, it remains a reliable and economical methodology.

It is interesting to note that due to the reproducibility of flexure tests, this approach has been adopted more widely in the USA to identify the Mode II interlaminar fracture toughness (G_{IIC}) through the end notch flexure (ENF) method as specified in ASTM D7905. However, due to friction between the two faces in this setup, unstable crack growth is often observed, preventing propagation toughness measurements. A closed feedback loop machine control system is required to ensure stabilisation. In contrast, the end-loaded split (ELS) methodology has notably been adopted in BS EN ISO 15114 in favour of the stable crack growth behaviour and is the preferred approach as outlined by the European Structural Integrity Society (ESIS). Recent research has shown the effectiveness of the experimental compliance (or compliance calibration, CCM) with has allowed for improved confidence in results due to the removed requirement of tracking the clamp position, Δ_{clamp} .

To conclude, the mechanical properties of UD laminate composite systems to this day continue to be widely debated as the adoption of composite materials continues to increase. However, while some tests have demonstrated improvements, these exploratory approaches are conducted on only one type of UD composite system (2,4). The primary contribution of this work is a comparative assessment of how standard mechanical test methods influence the measured properties and observed failure mechanisms of T300/914 carbon fibre reinforced epoxy matrix composite laminates. Mechanical property values obtained will also be validated with existing data where available.

2.0 Experimental Methodology

2.1 Material System and Testing Equipment

Toray TORAYCA® T300 is a polyacrylonitrile (PAN) based standard modulus carbon fibre, it offers a strong balance between cost, strength and stiffness, which makes it an attractive material for most composite use cases (5). The variant

used in this experiment is the 6k variant paired with the Hexcel HexPly® 914 matrix, as illustrated in Table 1.

Each lamina layer for this UD prepreg system after curing was ~0.125 mm. The layup orientation is listed in the heading of each test below. All tests were conducted on Instron 50kN 6800 series testing machines. Data acquisition⁶ was done primarily through the testing machine and recorded using Bluehill Universal developed by Instron.

Two types of strain gauges manufactured by Tokyo Measuring Instruments Lab were made available and used where required by the respective testing standard. Type FLAB-2-11 was used if axial direction (0°) was necessary only. If both axial (0°) and transverse (90°) directions were both of interest, Type FCAB-2-11 was used instead. The location of strain gauge attachment was prepared using a fine grit sandpaper and cleaned using IPA solvent and lint-free wipes. Next, strain gauges were then attached using a small amount of cyanoacrylate glue and soldered to pads that were attached in a similar way. Finally, a standard strain gauge test resistance value of 120 Ω was verified using a multimeter prior to the start of the test.

2.2 Tensile [0]₁₆

Tensile testing was conducted with most of the setup in accordance with CRAG RAE 300 with several modifications:

1. Axial and transverse strain from the front face of the specimen was captured as optically.
2. An extensometer was also attached but removed before fracture at around $P = 14$ kN.
3. Specimen width, $w = 10$ mm.
4. Specimen thickness, $t = 2$ mm.
5. Load speed, $v = 2$ mm/min.

Optical strain was captured using a slave camera and tracking software, illuminated using an LED matrix lamp. The setup used can be seen in Figure 1. Axial strain was captured by placing two Imetrum IM-ACC017 specimen marking bullseye stickers, approximately 10 mm away from the specimen end tabs within the gauge length. Transverse strain was captured from two white dots that were added using a white paint marker as indicated.

After ensuring vertical alignment, the parallel grips were then closed and a preload of 0.5 kN was applied before the readings

Table 1. Mechanical properties of composite system used.

Symbol	Parameter	Value	Source
$\sigma_{T,m}$	HexPly 914 Matrix Tensile Strength	47.7	(8)
$E_{T,m}$	HexPly 914 Matrix Tensile Modulus	3.9	(8)
$\varepsilon_{T,m}$	HexPly 914 Matrix Tensile Strain	0.015	(8)
$\sigma_{T,f}$	Toray T300 Fibre Tensile Strength	3530 MPa	(9)
$E_{T,f}$	Toray T300 Fibre Tensile Modulus	230	(9)
$\varepsilon_{T,f}$	Toray T300 Fibre Tensile Strain	0.015	(9)

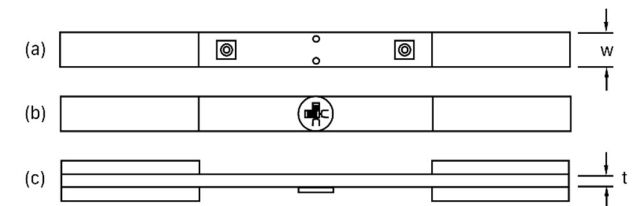


Figure 1. Diagrammatic representation of (a) front view with optical strain, (b) rear view with soldered strain gauges, and (c) non-tapered end tabs for tensile testing.

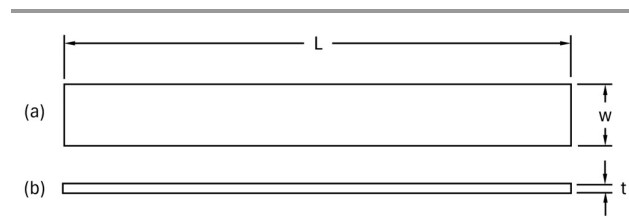


Figure 2. Diagrammatic representation of specimen (a) top view, (b) front view for flexure testing

were zeroed and test started. A total of eight specimens were tested due to the minimum requirement of five valid sets of data for the final values to be significant.

2.3 Flexural 3PT and Flexure 4PT [0]₁₆

Flexural testing was conducted with the setup in accordance with ASTM D790-17 and ASTM D6272-25 with minimal modifications.

1. Specimen width, w or $b = 25$ mm
2. Specimen thickness, t or $h = 2$ mm
3. Specimen length, $L = 150$ mm

A span size, S of 120 mm was used for both setups. Since thickness, t was 2 mm, a span-to-thickness ratio of 60 was calculated, corresponding to a load speed of 12 mm/min. Once the fixture was set up, the specimen, as pictured in Figure 2, was placed into the fixture and crosshead lowered before starting the test. Four specimens were tested for both configurations.

2.4 Compression [0]₁₆

Using CRAG RAE 400 and ASTM D3410M-16, a blended approach was used. While both share the same Celanese rig, the specimen design in CRAG was used as seen in Figure 3. A small preload of around 50 N was applied to reduce any compliance by aligning the faces within the fixture.

1. Specimen width, $w = 10$ mm
2. Specimen thickness, $t = 2$ mm
3. Specimen length, $L = 110$ mm

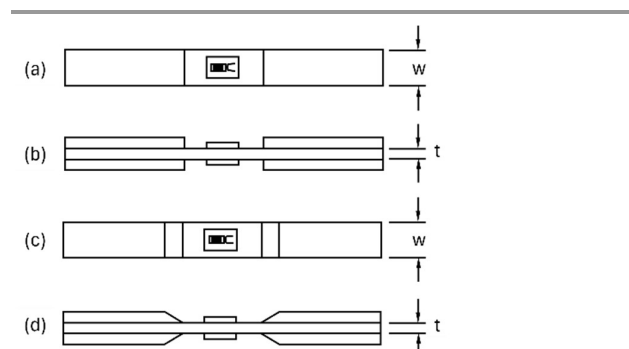


Figure 3. Diagrammatic representation of (a) CRAG specimen with 0° strain gauge on both sides, (b) CRAG specimen with tapered end tabs, (c) ICSTM specimen with strain gauges on both sides, (d) ICSTM specimen with non-tapered end tabs for compression testing.

On the other hand, the ICSTM method was conducted as specified in BS EN ISO 14126:2023 and (2). This method involved loosening the bolts that held the specimen to ensure the specimen was square within the fixture, after applying a preload of 500 N. This was held and the four bolts were tightened using a torque wrench from top to bottom in the following order: 10 Nm, 5 Nm, 5 Nm, and 10 Nm.

1. Specimen width, $w = 10$ mm
2. Specimen thickness, $t = 2$ mm
3. Specimen length, $L = 90$ mm

All preloads were applied using specimen protect mode. Prior to and during preload, strain gauge connections were also checked and calibrated. Loading speed was set at 1.5 mm/min for both setups. Four specimens were tested for each setup.

2.5 Shear

2.5.1 V-notched Beam (Iosipescu) [0]₂₄

Using ASTM D5379M-19e1, V-notched beam was manufactured to the desired specifications, as seen in Figure 4(a) and Figure 4(b), which was tested with the Wyoming test fixture. Two setups were manufactured, one with UD fibres parallel to the length of the specimen (IOSIPESCU 0), and another with UD fibres perpendicular (IOSIPESCU 90) to the length of the specimen.

1. Specimen width, $w = 12.5$ mm
2. Specimen thickness, $t = 3$ mm
3. Specimen length, $L = 75$ mm
4. Specimen $d_1 = 20$ mm

Each of these specimens was configured with a strain gauge that was rotated to form the +45 degrees and -45 degrees strains to calculate the shear strain. Four specimens were tested for each setup with a loading speed of 0.5 mm/min.

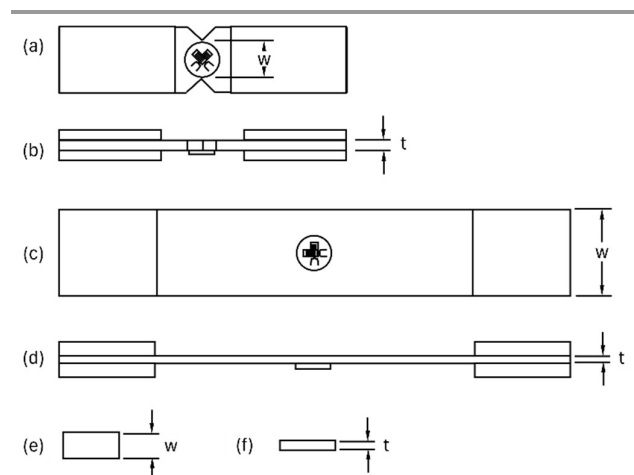


Figure 3. Diagrammatic representation of (a) Iosipescu specimen with strain gauge, (b) Iosipescu specimen with non-tapered end tabs, (c) ±45° tension specimen with strain gauge, (d) ±45° tension specimen with non-tapered end tabs, (e) and (f) short beam shear specimen.

2.5.2 $\pm 45^\circ$ Tension Test [$+45/-45/-45/+45$]s

This was manufactured in accordance to ASTM D3518M-18 as seen in Figure 4(c) and Figure 4(d). Strain gauges were attached using the method specified above. Likewise, four specimens were conducted at a loading speed of 2 mm/min.

1. Specimen width, $w = 25$ mm
2. Specimen thickness, $t = 1$ mm
3. Specimen length, $L = 150$ mm

2.5.3 Short Beam Shear [0]₁₆

This was manufactured according to ASTM D2344M-22, as seen in Figure 4(e) and Figure 4(f). The specimen was loaded in the fixture. Four specimens were tested at a speed of 1 mm/min.

1. Specimen width, $w = 10$ mm
2. Specimen thickness, $t = 2$ mm
3. Specimen length, $L = 20$ mm

2.6 Fracture

2.6.1 Double Cantilever Beam (DCB)

This was done in line with ASTM D5528. A thin polytetrafluoroethylene (PTFE) film was placed at the midpoint at 1.5 mm thickness of 3mm thick [0]₂₄ layup. This was inserted up till $a = 40$ mm and is the initial crack length prior to initiation. PTFE film provides a sharp tip and was left inside during the test.

The ASTM definition for calculating G_{IC} is used. Specimen dimensions are as follows:

1. Specimen width, $b = 20$ mm
2. Specimen thickness, $h = 3$ mm
3. Specimen length, $l = 150$ mm
4. End block $L' = 10$ mm
5. End block $t = 7.25$ mm

Once cut, end blocks for force exertion were glued on as seen in Figure 4, as such correction factor, F should be included for the reduced compliance. Crack length, force, and displacement values were recorded by eye using a travelling microscope for the crack propagation phase. Prior to this, a precrack was done by first clamping at 70 mm at 5 mm/min before unloading and reclamping at 100 mm and loading at 1 mm/min.

2.6.2 End-Loaded Split (ELS)

The ELS setup is based on BS EN ISO 15114. Like DCB, a thin PTFE film was placed at the 1.5 mm thickness between the lamina. This provided a sharp tip and was left inside during the test.

1. Specimen width, $b = 20$ mm
2. Specimen thickness, $2h = 3$ mm
3. Specimen length, $l = 150$ mm
4. End block $L' = 10$ mm
5. End block $t = 7.25$ mm

The ISO definition for calculating G_{IIC} is used in this experiment via the compliance calibration method. Similarly, a precrack was done by clamping at 70 mm first with loading speed 5 mm/min. Similar method was used by reclamping at 100 mm and loading at 1 mm/min. Crack length, force, and displacement values were recorded by eye using a travelling microscope for the crack propagation phase.

2.7 Summary

To summarise all the methodologies adopted and highlight where modifications were made, only CRAG standards were modified to reflect recent findings that have shaped modern testing methodologies. All the standards used have been summarised in Table 2.

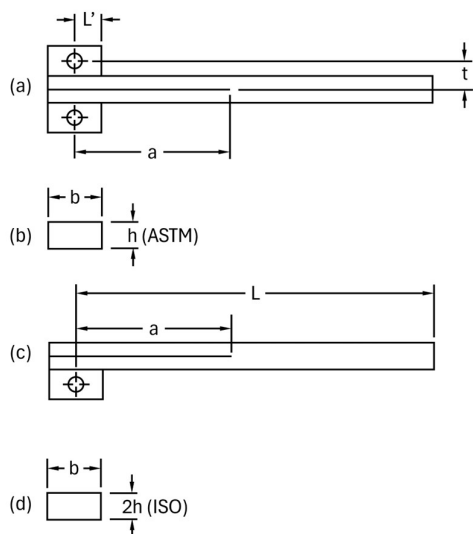


Figure 4. Diagrammatic representation of (a) DCB specimen side profile, (b) DCB cross section, (c) ELS specimen side profile, and (d) ELS specimen cross section.

Table 2. Summary of tests with corresponding setups and standards.

Test	Setup	Standard
Tensile	CRAG	CRAG RAE 300 [‡]
Flexural	3PT	ASTM D790-17
	4PT	ASTM D6272-25
Compression	CRAG	CRAG RAE 400 ^{‡‡}
	ICSTM	ASTM D6641M-23 ^{‡‡‡}
Shear	IOSIPESCU 0	ASTM D5379M-19e1
	IOSIPESCU 90	ASTM D5379M-19e1
	± 45	ASTM D3518M-18
	SBS	ASTM D2344M-22
Fracture	DCB	ASTM D5528-21
	ELS	BS EN ISO 15114:2014

[‡] Modifications made to CRAG RAE 300.

^{‡‡} Experimental procedure and data reduction methodology of ASTM D3410M-16 was employed instead.

^{‡‡‡} ICSTM findings informed ASTM D6641M-23. It should be noted that BS EN ISO 14126:2023 lists CRAG and ICSTM as valid methods.

3.0 Experimental Results and Discussion

3.1 Tensile

Results from the tensile test can be found in Table 3. Values for two approaches used to calculate the tensile modulus were considered in this approach. The 0.25% strain approach specified in CRAG was done by identifying the closest strain value to 0.25%, and then extracting the force for that value. This was divided by the gauge cross section area, A to calculate the tensile modulus, with Equation 1.

$$E_{2.5\%} = \frac{P}{0.0025 \cdot A} \quad (1)$$

Similarly, the chord modulus approach specified in ASTM (but also accepted in CRAG) was done with Equation 2.

$$E_{chord} = \frac{P_{3\%} - P_{1\%}}{0.002 \cdot A} \quad (2)$$

As specified in CRAG and ASTM, strain gauges are preferred for these calculations. With built-in displacement values, in most scenarios, the strain value calculated was consistently a factor of four larger than other strain values. A possible explanation for this could likely be the slippage between the end tab within the grip which relies on friction. Optical strain values were found to be comparable where $E_{2.5\%}$ was found to be 135.3 ± 5.6 GPa with a CV of 4.1% for five readings. With the extensometer, tensile modulus was found to be at 137.4 ± 8.3 GPa with a CV of 6%.

Despite both data acquisition methods achieving similar values, compared to strain gauges, optical strain extremely inconsistent in the transverse direction, with only two out of eight specimens returning data. Conversely, the extensometer is unable to track the ultimate tensile strength, σ_T as it was removed after $P=14$ kN. Through this, failure strains could also be approximated as seen in Figure 5.

Table 3. Tensile properties for T300/914 carbon/epoxy prepreg. S4 specimen was disregarded due to inavailability of specimen dimensions. S5 specimen was disregarded due to invalid failure mode with some failure occurring in the end tabs. This was expected due to the mismatched end tab lengths which likely resulted in applied force going off-axis. In addition, values for Poisson's ratio that were unusually small (<0.1) were also not considered. The average gauge cross section area of the specimens was 17.496 ± 0.754 mm², with a CV value of 4.31%.

Specimen number	$E_{2.5\%}$ (GPa)	E_{chord} (GPa)	σ_T (MPa)	ν
S1	144.5	146.6	1492	0.286
S2	143.0	145.9	1574	0.325
S3	135.3	137.8	1292	0.322
S4	x	x	x	0.278
S5	x	x	x	x
S6	147.6	149.5	1412	x
S7	150.6	153.4	1573	x
S8	131.8	134.3	1488	x
AVG	142.1	144.6	1472	0.303
SD	6.6	6.6	98	0.021
CV	4.7%	4.5%	6.6%	6.9%

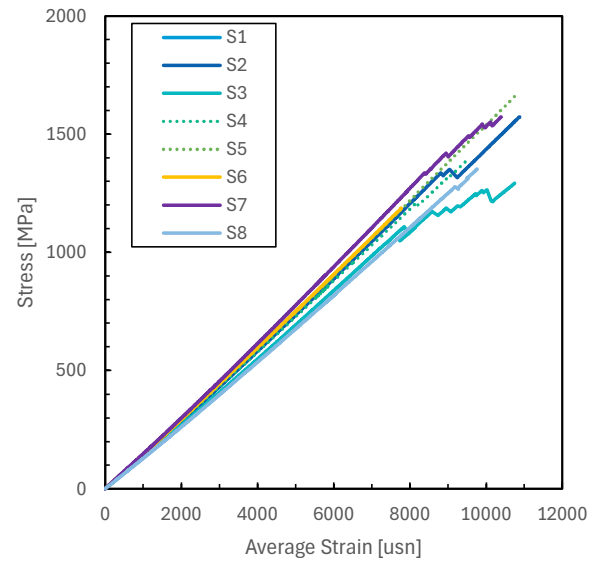


Figure 5. Stress-strain graph of T300/914 under tensile force. Dotted lines indicate invalid test specimens as described in Table 3. Significant variation on the failure strain (8000-12000 usn) makes it difficult to determine an accurate point for the 1% strain.

In a study conducted in (1: p.71) on T300/914 using ASTM D3039, σ_T was found to be 1535 ± 164 MPa whereas E_T was found to be 143.2 ± 10 GPa. The greater variability was noted to be driven by inhomogeneity from manufacturing the panels. As such, we can observe that similar values were attained in this study using CRAG RAE 300.

3.2 Flexural

Flexure stiffness was calculated using a similar approach to Equation 2. Upon finding the maximum stress value for the three-point beam setup (3PT), stiffness was calculated by the chord modulus method. Using this method, maximum flexure stress was found as 2097 ± 150 MPa with CV of 7.1%. Flexure stiffness was found to be 122.1 ± 3.8 with CV of 3.1%. All failure modes were valid with a clean split near the centre of the longitudinal direction.

On the other hand, four point (4PT) bending required corrected equation which accounted for the large deflections:

$$\sigma = \frac{PS}{bh^2} \left(1 + \frac{4.7D^2}{S^2} - \frac{7.04hD}{S^2} \right)$$

Where D stands for machine displacement. The stress strain graph seen in Figure 6, returns flexure stiffness of 150.9 ± 11.1 GPa with a CV of 7.4% and average maximum flexure strength is 1798 ± 26 (1.4%) MPa.

It is interesting to note that, flexural strength here with 4PT while avoids the issue of a concentrated load which creates a sharp transition in shear force, results in a lower strength. One hypothesis could be how the wider region that will be put under pure bending exposes a wider volume of the specimen to

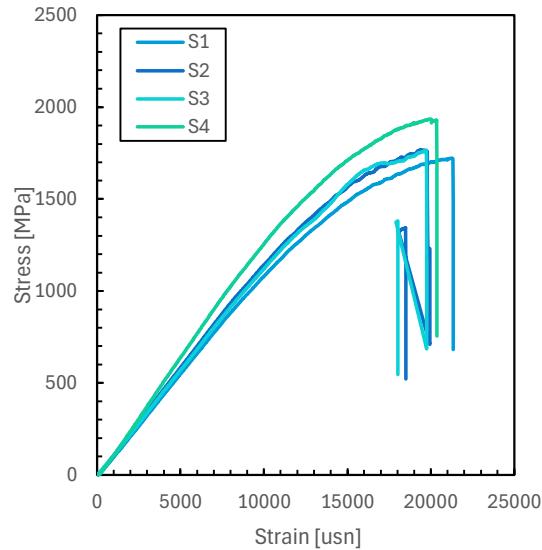


Figure 6. Stress-strain graph of T300/914 under four point bending. S2 and S3 have extremely similar curve profiles which highlights the repeatability of this experiment.

defects like porosity which may trigger failure, pointing to the Weibull effect also seen in fracture toughness.

3.3 Compression

With compression, one of the biggest conditions is alignment of the specimen within the fixture. With the support of dual strain gauges, percent bending allows checks to be performed on the validity of certain parameters of the test. By plotting bending against average strain as seen in Figure 7, easy identification when specimen values become invalid can be done. For compressive modulus, E_c values to be valid, the percent bending at 2000 usn must be below 10% as described in ASTM D3410M. Likewise, for ultimate compressive force to be valid, it is important to ensure bending percentage at failure strain also lies below the 10% axis. This should be critically evaluated objectively but also judged based on trendlines due to the effect of strain gauge failure which can result in values to be lower than actual. As such, careful evaluation of percentage bending is necessary.

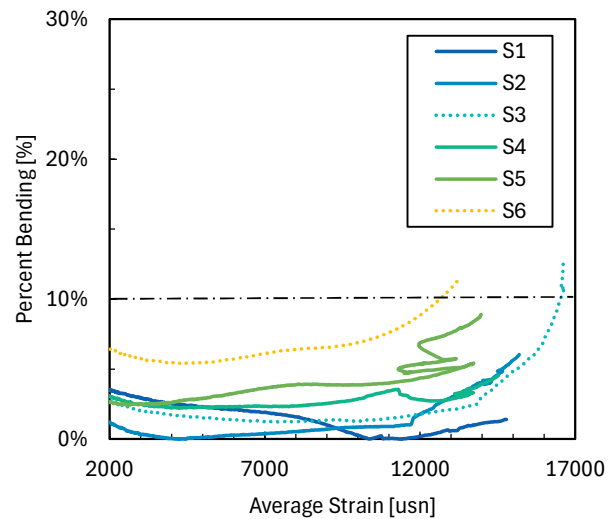
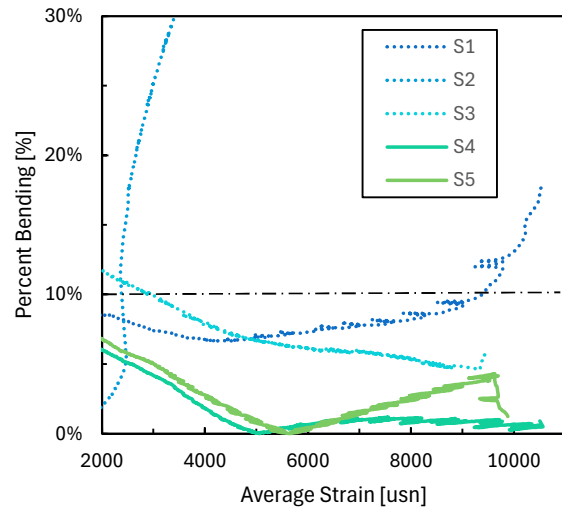


Figure 7. Percent bending plot for CRAG specimens (top) and ICSTM specimens (bottom), where dotted lines indicate specimens that failed to meet stay below 10%. CRAG specimens despite having valid failure modes, were less evenly loaded compared to ICSTM specimens.

Table 4. CRAG compression properties for T300/914 carbon/epoxy prepreg.

Specimen number	E_{chord} (GPa)	σ_T (MPa)	Failure mode
S1	148.0	1559*	BAT
S2	199.4*	1459*	BAT
S3	144.4	1396.4*	BGT
S4	132.4	1443	BAT
S5	137.0	1394	BGM
AVG	140.4	1419	All valid.
SD	7.06	34	
CV	5.0%	2.4%	

* Criteria for result validity not met, therefore value not included in calculation of metric. If failure mode not valid, E_{chord} and σ_C for specimen are ruled out.

Table 5. ICSTM compression properties for T300/914 carbon/epoxy prepreg.

Specimen number	E_{chord} (GPa)	σ_C (MPa)	Failure mode
S1	149.9	1774	HAT
S2	134.8*	1637*	TIT*
S3	144.7	1967*	BGT
S4	136.9	1657	DIB
S5	132.8	1572	HAT
S6	129.5	1476.8*	HAT
AVG	138.8	1668	S2 invalid.
SD	8.4	102	
CV	6.1%	6.1%	

* Criteria for result validity not met, therefore value not included in calculation of metric. If failure mode not valid, E_{chord} and σ_C for specimen are ruled out.

With regards to the results from this experiment, as seen in Table 4 and Table 5, while compressive modulus is approximately equal, the compressive strength is about 200 MPa higher using the combined shear and end loading methodology in ICSTM. This mirrors the results observed in the original proposal for the ICSTM compression fixture which was conducted for XAS/914C carbon/epoxy UD composite (2).

As indicated by the bending percentage, the reason for this increased stress loading comes down to how well aligned and how constrained the specimens were. As highlighted by Matthews, even slight changes in specimen clamping resulted in up to 30% of the measured compressive strength (1: p.93), which is a variable that the ICSTM jig achieves through torque control of the specimen clamping force.

3.4 Shear

In-plane shear strength (IPSS) and interlaminar shear strength (ILSS) are key parameters in understanding the shear properties of UD composite laminates. In this section, key properties from each test have been summarised in Table 6. The analysis conducted were in accordance to their respective standards.

From the results, we can identify that while the Iosipescu V-notch approach has led this area due to its ability to approach the theoretical limit, the IOSIPESCU 90 setup comes with significant variation (6). This can be attributed to the fragile nature of this specimen which as noted in ASTM D5379M underestimates IPSS values by 20%. Perhaps interestingly, [0/90]_{ns} systems is recommended to be most accurate by ASTM.

This underestimation has been suggested to be due to the uniform shear stress state assumption between the opposing

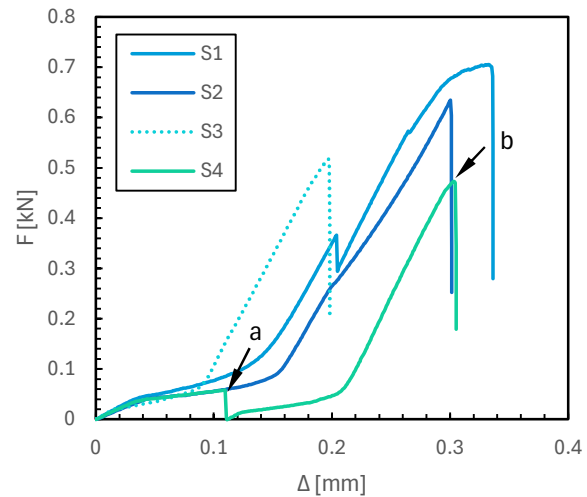


Figure 8. Force displacement curve of IOSIPESCU 90 specimens. Exhibit (a) indicates a premature failure which results in the force dropping. Exhibit (b) follows up to this showing the peak shear force applied to be significantly lower for S4.

notches. However, other considerations include force eccentricity and notch geometry. As stated in the ASTM D5379M: “fixture tolerancing, unstable or thin geometry, improperly installed in the fixture, out-of-tolerance because of poor specimen preparation, or of a material configuration with an extremely low tolerance to twist.” As a result, it is difficult to exactly pin-point the cause for the uncertainties. Recommendations for the future include the need to consider a [0/90]_{ns} approach to shear testing, and the installation of dual-side strain gauges to monitor specimen twisting.

On the flip side, ASTM D3518 ±45 tension test is a well established method to determine shear properties. The only

Table 6. Shear properties of T300/914. For IOSIPESCU 90, S3 had an invalid failure mode with some failure within the end tab, in addition, S4 strain gauge values significantly deviated (negatively) after 24.6 seconds, as a result, these values were excluded.

	Shear Modulus (GPa)			
	0deg	90deg	±45	SBS
AVG	5.95	3.08	5.10	-
SD	0.31	1.76	0.17	-
CV	5.3%	57%	3.3%	-
	Ultimate Shear Strain (usn)			
	0deg	90deg	±45	SBS
AVG	61765	7190*	-	-
SD	9676	5057	-	-
CV	15.7%	70.3%	-	-
	Ultimate Shear Strength (MPa)			
	0deg	90deg	±45	SBS
AVG	102.79	17.02	92.34	74.73
SD	4.19	2.14	0.47	2.28
CV	4.1%	13.0%	0.5%	3.1%

* Only two specimens were used in determining these values. Outside of these, all other testing systems had at least three specimens.

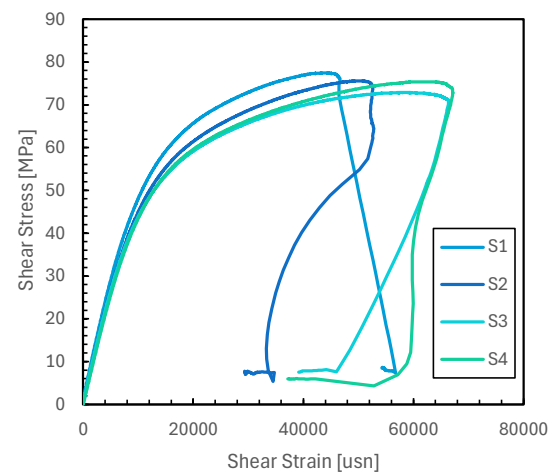


Figure 9. Shear stress to shear strain plot for ±45 tension test with consistent shear modulus and ultimate strength values. Note that 5% shear strength was taken instead of the maximum for S3 and S4 due to the value exceeding 50000 usn.

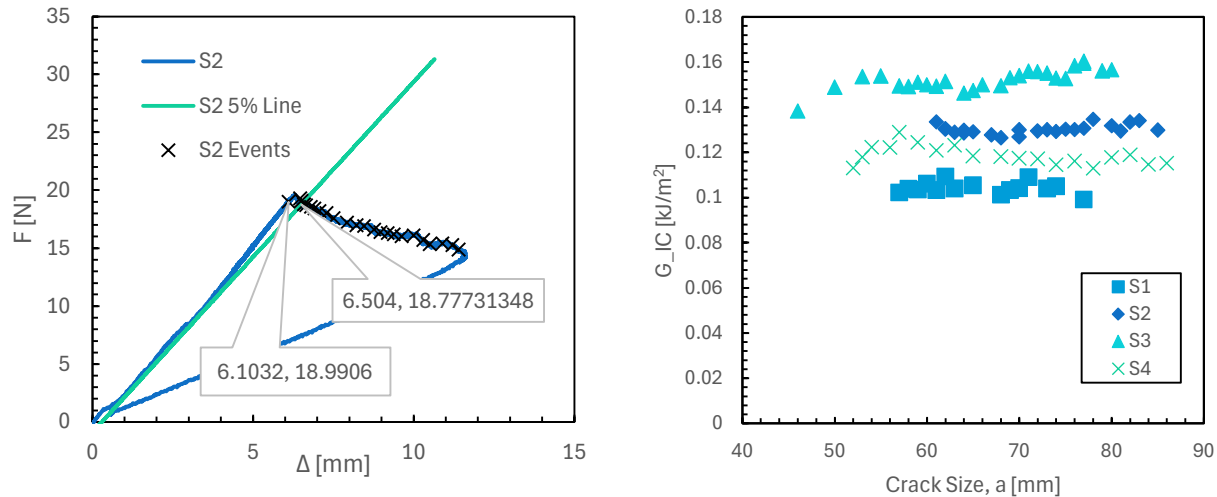


Figure 10. Force-displacement diagram of T300/914 for Mode I fracture toughness (left). P_{max} and $P_{5\%}$ values indicated using call outs. As $P_{5\%}$ is after P_{max} , therefore P_{max} should be used to calculate G_{IC} initiation. R-curve for T300/914 carbon/epoxy UD laminate (right) with propagation G_{IC} values plotted.

downside here is the inability to determine ultimate shear strain. In exchange, the CV value and consistency that comes with this methodology can be extremely beneficial in material development, as seen in Figure 9.

3.5 Fracture

Double cantilever beam (DCB) method requires careful consideration on the method used to calculate fracture toughness. The modified beam theory (MBT) and compliance calibration method (CCM) are two main data reduction methods. In recent years, CCM has largely replaced MBT, however, as noted in BS EN ISO 15114:2014, some solutions for MBT still remain.

In DCB specimens tested, most cracks were conducted successfully with a dull crack. However, while analysing the precrack data, G_{IC} precrack initiation was found to average around 0.167 ± 0.127 kJ/m² with a CV of 76%. This value was calculated by taking P_{max} from the precrack force-displacement diagram. The large variance in values likely to do with how well embedded the film is within the specimen.

After precracking, G_{IC} initiation was computed to be an average of 0.127 ± 0.014 kJ/m² (CV = 11.1%). This was done using the method highlighted in Figure 10. As the T300/914 prepreg laminate used did not have any strengthening mechanisms, G_{IC} propagation was found to also be quite similar at 0.126 ± 0.0203 kJ/m² (16.1%). This was achieved by averaging the G_{IC} values after initiation. A comparative assessment was also conducted where both MBT and CCM values were compared for Mode I and the difference in values across the board was found to be less than 0.01 kJ/m² when correction factors were taken into account.

With regards to the end-loaded split approach, the initiation precrack G_{IIC} was calculated to be 1.08 ± 0.163 (15%) kJ/m². G_{IIC} initiation was also calculated as 0.343 ± 0.064 (19%) kJ/m² with a similar method in DCB. G_{IIC} propagation averaged out to 0.324 ± 0.0711 (22%) kJ/m². Interestingly, for specimen S3, a shiny surface was observed after the crack ended, indicating a slow but stable crack growth. This explains the values that have been calculated here, which are around two to three times higher than Mode I.

As seen in both Figure 10 and Figure 11, the R-curves for both Mode I and Mode II stays relatively flat. This is something that can be expected due to the lack of toughening mechanisms like Z-pinning, and thermoplastic interleaf (7). The use of a relatively weak epoxy matrix in HexPly 914 compared to PEEK for instance explains the values observed in this test.

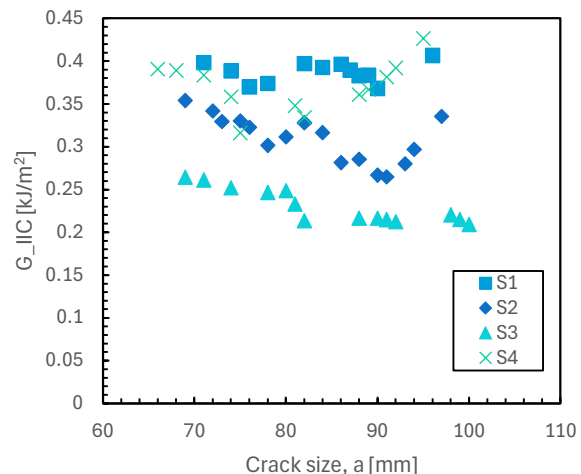


Figure 11. R-curve for Mode II fracture toughness G_{IIC} .

Conclusion

This study attempted to quantify the changes between various standards and approaches by exploring the mechanical properties of UD laminates. The mechanical properties for T300/914 were tested with some modifications based on ASTM, CRAG, and ISO. Crucial findings in this process include the effectiveness of these standards on a wide number of composites, of which T300/914 was investigated in this study. Most notably, in compression testing, the advantage of a combined shear and end loading fixture was observed. Where an improvement similar to the findings first published were replicated successfully with T300/914. Future work is necessary to ensure gaps in this study can be addressed. One area for potential further research would be quantifying the benefits of using a [0/90]_{ns} arrangement when conducting Iosipescu shear testing.

Acknowledgements

The author would like to thank the support provided by the technicians at the The Composites Centre, Imperial College London for assisting with manufacturing the specimens required for this study.

Notes and references

§ Data collected and presented in this study was collected across separate testing sessions performed by independent groups. The resulting dataset will demonstrate the differences between laboratories when implementing the same standard. The variability in data collected reflects the practical variability that arises when different laboratories implement the same standard.

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GENERAL RISK ASSESSMENT

SUBJECT OR ACTIVITY:- Mechanical Testing of Polymer Composites

1. PERSONS CARRYING OUT, APPROVING AND CHECKING THIS ASSESSMENT. (MUST HAVE SUCCESSFULLY COMPLETED RAFT OR OTHER APPROVED TRAINING)

Assessor	Zhe Wei Kho	Position	MSc Composites Student	Date	28/11/2025	Confirm Training in Risk Assessment	Yes ☒
Approved by	Dr Emiliano Bilotti	Position	MSc Composites Director	Date	28/11/2025	Confirm Training in Risk Assessment	Yes ☒
Checked by	Avishka Jayasinghe Arachchi Keith Wolstenholme	Position	Lab Technician Lab Manager	Date	28/11/2025	Confirm Training in Risk Assessment	Yes ☒

2. DETAILED DESCRIPTION OF THE STEPS INVOLVED IN THE ACTIVITY (include storage, transport and disposal if relevant, and how often it is carried out)

This risk assessment covers the execution of five different types of mechanical tests, namely Tension, Compression, Shear, Flexure and Fracture tests of carbon fibre-reinforced polymer composites. The mechanical tests are performed in accordance with their respective test standards, namely: Tension – ASTM D3039-17, Compression – ASTM D3410-16 & ASTM D6641-16, Shear – ASTM D3518-18 & ASTM D5379-19 & ASTM D2344-16, Flexure – ASTM D790-17 & ASTM D6272-17, and Fracture – ASTM D5288-13 & BS ISO 15144-2014.

Conducted as groups of four or five students, laboratory technicians and graduate teaching assistants (GTAs) will be assisting and monitoring throughout the test campaign. These are conducted using an Instron 50kN test machine. Before starting, specimens are prepared through demonstrations with students and hands-on opportunities for experience. Each equipment (if applicable) is operated in accordance with its respective risk assessment. After an initial briefing, the machine is turned on and each composite test specimen is then loaded by the technicians.

Prior to the test starting, students are instructed to stay away from the machine and reminded to have safety goggles on. Once the debris shield(s) are in place, the experiment is then initiated and data is recorded automatically as well as by the students throughout this process. This information collected will be used for additional analysis to obtain information on properties such as flexural stiffness, tensile stiffness, compressive stiffness, tensile strength, and many others.

Upon completion of the experiment, a discussion is conducted with the laboratory technicians and graduate teaching assistants. Students are asked to examine the fracture surfaces, assess the validity of the failure mode, understand the load cases that occurred, and determine possible causes for failure. Additional time is provided for thorough discussion and for students to ask questions.

3. LOCATION

Site	SK	Building	CAGB	Room	002
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4. HAZARDS – WHAT ARE THEY, WHO IS AFFECTED, HOW ARE THEY CONTROLLED?

NOTES – PLEASE READ CAREFULLY!

List **all** the hazards associated with the work, decide who could be affected, give details of what could happen without any controls, and estimate the “raw” risk.

Then write down the control measures that are already in place to prevent any harm occurring, and estimate the “residual” risk using the matrix guide (Appendix1)

If the residual risk is not reduced to “LOW” (less than 4) then additional controls must be considered; however, aim to get the lowest risk score that is reasonably practicable.

HAZARD TYPE	WHO DOES THE HAZARD AFFECT?	WHAT INCIDENT OR HARM COULD BE CAUSED WITHOUT ANY CONTROL MEASURES?	RAW RISK with no control measures. In your estimation, is it:- LOW, MEDIUM or HIGH?	EXISTING CONTROL MEASURES (E.g. PHYSICAL THINGS LIKE MACHINE INTERLOCKS; PROCEDURAL THINGS LIKE OPERATING PROCEDURES; TRAINING)	RESIDUAL RISK with existing controls (See appendix 1)		
					Severity	Probability	Risk (S x P)
Sharp Objects	Staff / Students	Sharp edges on test specimen due to imperfections can cause cuts. Loose fibres on composite specimens can also cause splinters which can cause irritation over time if left unattended.	LOW	Careful handling of test specimens, use pliers, or wear gloves if appropriate.	1	4	4
Mechanical	Staff / Students	Moving machine parts could result in physical crushing injuries or broken bones due to trapping or impact. Flying debris from test specimens could result in cuts, impact injuries or eye injuries.	MEDIUM	Operator training to be carried out and documented before use, safety guarding in place, safety glasses, and overalls to be worn whilst setting and operating the machine. Specimen protect should be turned on and limit stops adjusted to prevent crushing injuries	3	1	3
Electrical	Staff / Students	Shocks from mains supply voltage to the machine and controller could potentially be fatal. Cables used with any instrumentation devices may also pose electrical hazards. The emergency safety stop relies on the correct operation of the machine wiring system.	HIGH	Insulation of mains supply, regular visual checks for damage, trailing leads to be securely held down to prevent any potential trip hazards or damage to cables. Before setting up a test the emergency stop should be checked for correct operation.	1	4	4

Manual handling	Staff / Students	Handling of heavy apparatus and testing equipment could cause strain, sprain, or serious spinal injuries.	MEDIUM	Fixtures and fittings should be assembled and removed by trained technical staff to prevent accidental injuries. Heavy equipment should be moved around using wheels	3	1	3
Trapping	Staff / Students	Moving parts could potentially trap hands when setting up the machine, specifically crosshead and grips can be overextended, causing injuries or accidental loading of specimen.	MEDIUM	Use specimen protect and set the safety stop limits when moving the crosshead. Take care when opening and closing grips/fixtures. Ensure lab coats are worn to prevent loose clothing.	3	2	2
Falling, slips and trips	Staff / Students / Other	Loose, untidy floorspace could result in broken bones or serious spinal or head injuries from tripping over.	LOW	Make sure there are no trailing leads or objects left lying around the work area. Make sure tools and associated test equipment is stored in cupboards, drawers, and racking. Clean up any spills and use portable safety barriers to control any access where appropriate.	3	1	3

5. WHAT ADDITIONAL CONTROL MEASURES ARE NEEDED?

If none, please tick box and move on to next section

☐

Please describe:-	Responsible Person(s)?	By what date?	Completed? (Date)
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.	Click or tap to enter a date.
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6. LONE WORKING (This can increase the risk from other hazards, and must always be considered.)

NOTE – READ CAREFULLY! USE THESE LINKS FOR THE [COLLEGE GUIDANCE](#), THE COLLEGE [CODE OF PRACTICE](#) AND THE [LONE WORKING CONSENT FORM](#)

Lone working is not a hazard in itself, but will increase the risk of harm occurring from other hazards.

Record *why* Lone working will or won't take place, state *who* will be lone working and *what* controls and permissions are in place.

WHY WILL (OR WON'T) IT TAKE PLACE?	WHO MIGHT BE WORKING ALONE?	WHAT ARE THE CONTROLS / PERMISSIONS?
Lone working in the Dynamic Fracture and Forming (DFF) Lab is not permitted.	Click or tap here to enter text.	Click or tap here to enter text.

7. EMERGENCY ACTIONS

INCLUDE:-

- Dangerous failure of equipment or experiment;
- Evacuation procedures if the (Fire) Alarm sounds (Evacuation route and muster point);
- How to raise the alarm;
- What to do and who to call in the event of an Injury or accident.

Upon the sounding of the fire alarm, the following emergency procedures should be conducted:

1. Immediately stop all experiments and depress the emergency stop button for the Instron 50kN
2. Exit the lab
3. Leave via the defined route detailed in the induction
4. In the event of a medical emergency, call 4444 (Imperial College Campus Security). State the location and nature of the incident.
5. Call technical supervisor: Keith Wolstenholme (Extn. 45060) or Suresh Viswanathan (Extn. 50380, Mobile: 07748350636)

Evacuation procedures (route and muster point)

1. Exit the lab
2. Walk towards emergency exit and congregate at Queen's Tower Lawn

Fire Prevention:

1. Immediately stop all experiments and depress the emergency stop button
2. Exit the lab, access fire type.
3. Access fire extinguisher and attempt to extinguish if appropriate by pulling the pin, aiming and exerting force on handle

8. Monitor and review

This risk assessment should be reviewed monthly until the extra controls are fully implemented, and annually thereafter.

It should also be reviewed Immediately in the event of process / location change or incident or accident.

[illegible]

9. Other Specialised Risk Assessments

Some of your hazards may require additional risk assessments - ask your safety officer for guidance or visit the [Central Safety forms web pages](#). For example:-

Chemical and Hazardous Substances – needs a [COSHH assessment](#).

Biological Hazards – visit the [Biological Safety](#) web pages – may need a College Bio1 risk assessment

Off-site Work and Fieldwork – there are [specific forms](#) for Fieldwork, Hosted Research and Conferences etc.

Ionising Radiation, inc. X-rays – the College Radiological Protection Team will help, but you need to [register the work](#).

Lasers - need to be registered and risk assessed

Significant Manual Handling – Speak to your departmental manual handling advisor

Compressed Gases and Cryogenics – need separate risk assessments.

PLEASE LIST BELOW ANY SPECIALISED RISK ASSESSMENTS THAT ACCOMPANY THIS ONE.

[illegible]

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Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.

10. Standard Operating Procedure (SoP)

This describes the full method by which the experiment or activity takes place. Please include all steps, thinking about setup, running, packing up, disposal etc
You may wish to provide a separate SoP. If so, please indicate this below and leave this section blank.

1. Tension Test (ASTM D3039-17)

System Setup

- Ensure the Instron 50 kN frame is powered on and has completed its internal checks.
- Install the appropriate **mechanical wedge grips** or hydraulic grips specified for composite coupons.
- Verify load cell calibration and zero the extensometer (if used).
- Confirm debris shields are functional and can be closed securely.
- GTAs/technicians check crosshead speed settings per ASTM D3039.

Specimen Preparation

- Inspect each tensile coupon for dimensional accuracy, tab alignment, and surface defects.
- Measure width, thickness, and gauge length using calibrated tools; record values.
- Attach strain gauges or clip-on extensometers if required for modulus measurement.
- Ensure tabs are properly bonded and cured (if applicable).

Running the Experiment

- Technician loads the specimen into the grips, ensuring alignment to avoid bending.
- Students stand clear; safety goggles on; debris shield closed.

- Zero load and strain channels.
- Start the test at the prescribed crosshead rate.
- Students record load–displacement data, failure mode, and any audible/visual cues.
- Machine stops automatically at specimen fracture.

Packing Up

- Remove broken fragments using gloves and tongs.
- Clean grip faces of debris or tab residue.
- Reset crosshead to a safe height for the next specimen.

Disposal

- Place fractured composite pieces in designated composite-waste bins.
- Dispose of strain gauges, backing paper, and tabs in general waste unless contaminated with adhesives.

2. Compression Test (ASTM D3410-16 & ASTM D6641-16)

System Setup

- Install the appropriate compression fixture (e.g., IITRI fixture or Combined Loading Compression fixture).
- Verify fixture alignment using manufacturer alignment rods.
- Check load cell zero and ensure platen surfaces are clean and parallel.

Specimen Preparation

- Inspect specimens for squareness, correct thickness, and absence of delamination.
- Measure dimensions and record.
- Apply strain gauges if modulus data is required.
- Ensure end tabs (if used) are properly bonded.

Running the Experiment

- Technician inserts the specimen into the fixture, ensuring correct clamping torque.
- Students step back; debris shield closed.
- Zero load and strain channels.
- Apply compressive load at the standard-specified rate.
- Students observe buckling behaviour, load drop, and failure mode.
- Test ends automatically at failure or displacement limit.

Packing Up

- Release fixture clamps and remove specimen remnants.
- Clean fixture surfaces and remove debris.
- Reset crosshead.

Disposal

- Place crushed composite fragments in composite-waste bins.
- Dispose of strain gauge wires and adhesive waste appropriately.

3. Shear Test (ASTM D3518-18, D5379-19, D2344-16)

(Covers $\pm 45^\circ$ tension shear, V-notched beam shear, and short-beam shear)

System Setup

- Install the correct shear fixture (e.g., V-notched rail shear fixture or short-beam three-point fixture).
- Confirm alignment and fixture tightness.
- Zero load cell and displacement channels.

Specimen Preparation

- Inspect for correct fibre orientation ($\pm 45^\circ$ for D3518), notch quality (D5379), or span-to-thickness ratio (D2344).
- Measure thickness, width, and gauge length.

- Apply strain gauges if shear strain measurement is required.

Running the Experiment

- Technician loads the specimen into the fixture, ensuring correct orientation.
- Students step back; debris shield closed.
- Start test at the prescribed rate.
- Students record load–displacement behaviour and note shear-related failure patterns (e.g., matrix cracking, delamination).
- Machine stops at failure or displacement limit.

Packing Up

- Remove specimen remains and clean fixture surfaces.
- Inspect fixture for wear or notch residue.
- Reset crosshead.

Disposal

- Dispose of fractured shear coupons in composite-waste bins.
- Remove and discard any loose fibres or splinters safely.

4. Flexure Test (ASTM D790-17 & ASTM D6272-17)

(Three-point or four-point bending)

System Setup

- Install the bending fixture with the correct span length per the standard.
- Ensure rollers are clean, parallel, and rotate freely.
- Zero load and displacement channels.

Specimen Preparation

- Inspect specimens for flatness, correct thickness, and absence of warping.
- Measure width, thickness, and span length.

- Mark mid-span if required for deflection measurement.

Running the Experiment

- Technician places the specimen on the supports.
- Students step back; debris shield closed.
- Apply load at the standard-specified rate.
- Students observe deflection, cracking, and failure mode (compression top-surface failure, tension bottom-surface failure).
- Test ends at fracture or at a defined strain limit.

Packing Up

- Remove broken pieces and clean rollers.
- Reset span length if switching between specimen types.
- Return crosshead to a safe height.

Disposal

- Place fractured flexural specimens in composite-waste bins.
- Remove any loose fibres from the test area.

5. Fracture Test (ASTM D5528-13 & BS ISO 15144-2014)

(Mode I DCB, Mode II ENF, or mixed-mode tests)

System Setup

- Install the fracture-mechanics fixture (DCB hinges, ENF supports, or mixed-mode loading blocks).
- Verify alignment and hinge pin engagement.
- Zero load, displacement, and crack-opening displacement (COD) gauges.

Specimen Preparation

- Inspect pre-crack insert (PTFE film) for correct length and placement.
- Measure thickness, width, initial crack length, and record.

- Apply crack-length scale markers along the specimen edge.
- Attach clip-gauge or optical measurement system if required.

Running the Experiment

- Technician mounts the specimen carefully to avoid altering the crack front.
- Students step back; debris shield closed.
- Apply load at the prescribed rate, ensuring stable crack growth.
- Students record load, displacement, crack length, and observe fracture propagation.
- Test ends when crack reaches the required length or catastrophic failure occurs.

Packing Up

- Remove specimen halves and collect any splintered fragments.
- Clean fixture surfaces and remove adhesive residue from hinges if present.
- Reset crosshead and measurement devices.

Disposal

- Place fractured DCB/ENF specimens in composite-waste bins.
- Dispose of PTFE inserts, tape, and gauge adhesives in general waste.

APPENDIX 1

Category	SEVERITY (s) Examples	Score
MINOR	Superficial injuries - cuts, bruises, mild skin irritation, mild aches and pains – requiring first aid only. Minor property damage.	1
SERIOUS	More serious injuries or ill-health, requiring time off work or study or a hospital visit, e.g. burns, sprains, strains and short-term musculoskeletal disorders, cuts requiring stitches, back injuries, fractures to fingers or toes. More serious property damage.	2
MAJOR	Broken limbs, amputations, long-term health problems resulting from work, or acute illness requiring medical treatment, loss of consciousness, serious electric shock, loss of sight. Major property damage.	3
FATAL	Injury or ill-health which leads to death either at the time or soon after the incident, or eventually, as in the case of certain occupational diseases, such as asbestos-related cancers.	4

- These tables are designed to help you to gauge whether your risks are low or high.
- However, there are many factors that affect one's judgment of risk – for example, you may not have all the information you need to make a realistic assessment. Lack of space, lack of training, working to deadlines, and working late and alone could all act as risk increasing factors.
- Your assessment of risk will always be subjective, as it depends on not only your knowledge and experience but also the information you have available. The more information you have, the more accurate the assessment will be.
- Remember, any effective, appropriate control measures will lower risk.

Category	PROBABILITY (p) The <i>likelihood</i> of the hazard causing harm (example for guidance only - some or all may apply for each category)	Score
VERY UNLIKELY	Good control measures are in place. Controls do not rely on a person using them (i.e. personal compliance). Controls are very unlikely to break down. People are very rarely in this area or very rarely engage in this activity.	1
UNLIKELY	Reasonable control measures are in place but they do rely on a person using them (some room for human error). Controls unlikely to breakdown. People are not often in this area / do not often engage in this activity / this situation is unlikely	2
POSSIBLE	Inadequate controls are in place, or likely to breakdown if not maintained. Controls rely on personal compliance. People are sometimes in this area or sometimes engage in this activity / this situation sometimes arises	3
LIKELY	Poor or no controls in place. Heavy reliance on personal compliance (lots of room for human error). People are often in this area / engage in this activity on a regular basis / this situation often arises.	4

Risk Level	
Low	
Medium	
High	
Very high	

RISK SCORE = S X P		SEVERITY OF OUTCOME (S)			
		Minor	Serious	Major	Fatal
P R O B A B I L I T Y (P)	Very Unlikely	1	2	3	4
	Unlikely	2	4	6	8
	Possible	3	6	9	12
	Likely	4	8	12	16